CS772: Deep Learning for Natural Language Processing (DL-NLP)

Introduction

Pushpak Bhattacharyya
Computer Science and Engineering Department
IIT Bombay

Week 1 of 28th July, 2025

Course Info

Website is important

- https://www.cse.iitb.ac.in/~cs772
- Channels of communication: MS Teams, Moodle, Course Website
- Also, visit moodle frequently

Evaluation Scheme (tentative)

- Quizzes- 2 nos., 15% each
- Thursday just before midsem and endsem
- No midsem and endsem
- 4 Assignments (40%)
- Challenging Course Project (30%)
- Programming Heavy!!

Assignments and Project

- Continuous evaluation
- Meeting every two weeks to monitor progress
- Credit for thorough literature survey for the project work

Journals and Conferences

- Journals: Computational Linguistics, Natural Language Engineering, Journal of Machine Learning Research (JMLR), Neural Computation, IEEE Transactions on Neural Networks

- Conferences: ACL, EMNLP, NAACL, EACL, AACL, NeuriPS, ICML

Useful NLP, ML, DL libraries

- NLTK
- Scikit-Learn
- Pytorch
- Tensorflow (Keras)
- **Huggingface**
- Spacy
- Stanford Core NLP

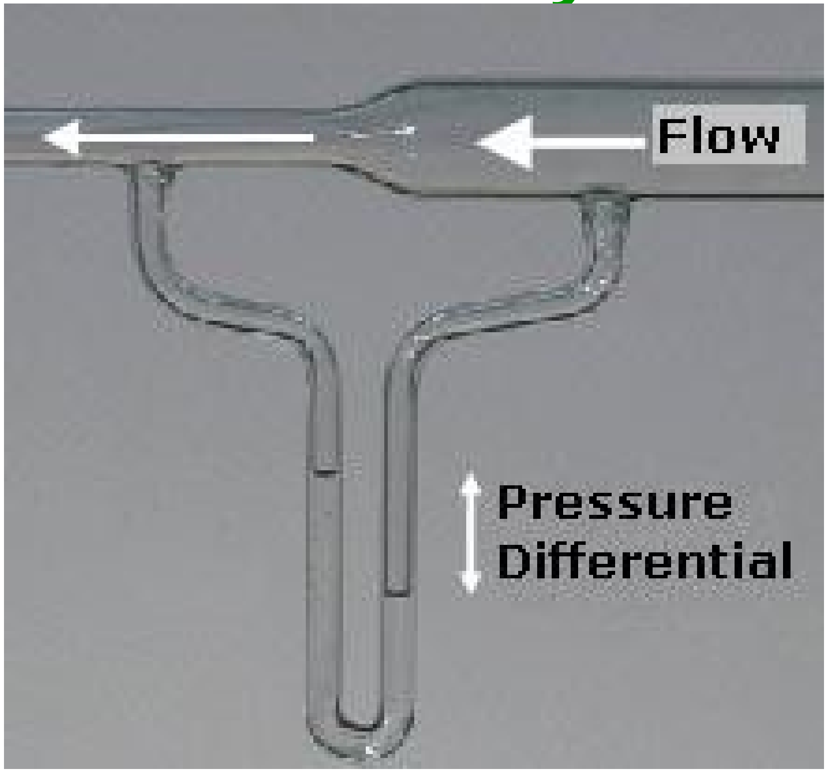
Natural Language Processing of Rare Language Phenomena and in Low Resource Setting

Ode to *Scientists and Engineers*

Scientists ask WHY
Engineers ask WHY NOT
Scientists wonder at WHAT-IS
Engineers wonder WHAT-COULD-BE
World couldn't do without either.

Scientists STUDY
Engineers MAKE
And ever the twain shall meet.

Phenomena and Modelling: Physics shows the way



Phenomenon

an increase in the speed of a fluid occurs simultaneously with a decrease in static pressure

Model →

$$\frac{v^2}{2} + gz + \frac{p}{\rho} = constant$$

Bernoulli's Principle

v- velocity
z- height
p= pressure
ρ- density

Perspective-1: Language

What is Language *(George Yule, "Study of Language", 1998)*

- **Displacement** (Indicators that change with time and place: *I saw him yesterday at the market; I will see him tomorrow in the school*)
- **Arbitrariness** (name → Meaning; *water, chair*)
- **Productivity/creativity** (potentially infinite no. of sentences)
- **Cultural Transmission** (child acquires parent's language)
- **Discreteness** (sound and meaning units separated)
- **Duality** (Surface structure, deep structure)

What is “language phenomenon”

- Sandhi (phonetic phonological transformation at morpheme boundaries)
  - गुरु + आदेश = गुर्वादेश: *guru* + *aadesh* = *gurvaadesh (The order from the teacher)*
- Fronting of verbs in question formation in German
  - *Trinkst du Tee? (Do you drink tea)*
- Semantic/Pragmatic Incongruity leading to Irony, Sarcasm
  - An irate unattended invitee leaving the party and being asked *“how did you enjoy the party?”* and replying *“O, I love being ignored”*

Perspective 2: NLP

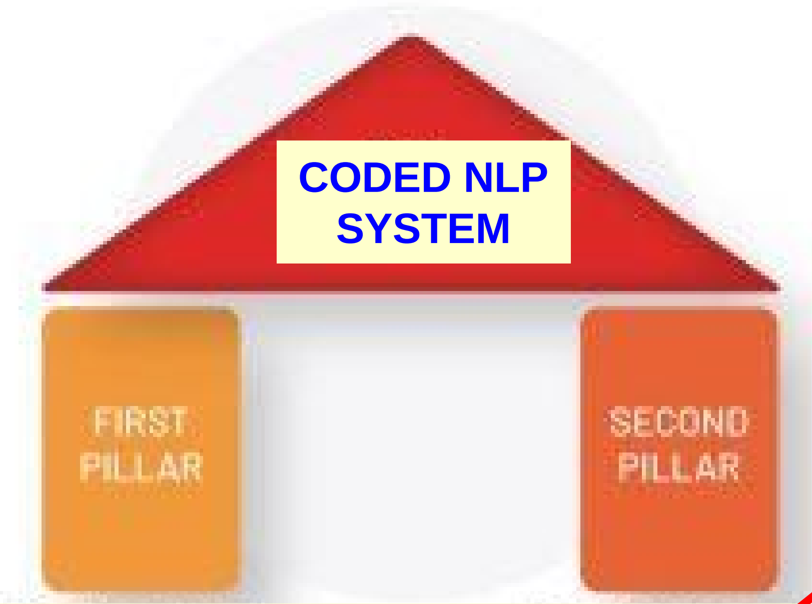
Natural Language Processing: NLU and NLG

- 3 Generations
  - Gen1- Rule based NLP is also called Model Driven NLP
  - Gen2- Statistical ML based NLP (*Hidden Markov Model, Support Vector Machine*)
  - Gen3- Neural (Deep Learning) based NLP
  - Gen3.5- LLM and GenAI

Two Pillars of NLP

BEFORE the model operates: *Annotation*

AFTER: Error Analysis



Central Limit Theorem

Law of Large Numbers

LINGUISTICS + PROBABILITY

= NLP

Main challenge is Ambiguity!: an extreme example

"Buffalo buffaloes Buffalo buffaloes buffalo buffalo Buffalo buffaloes"

Prompt to chatGPT: what do you understand by the above sentence

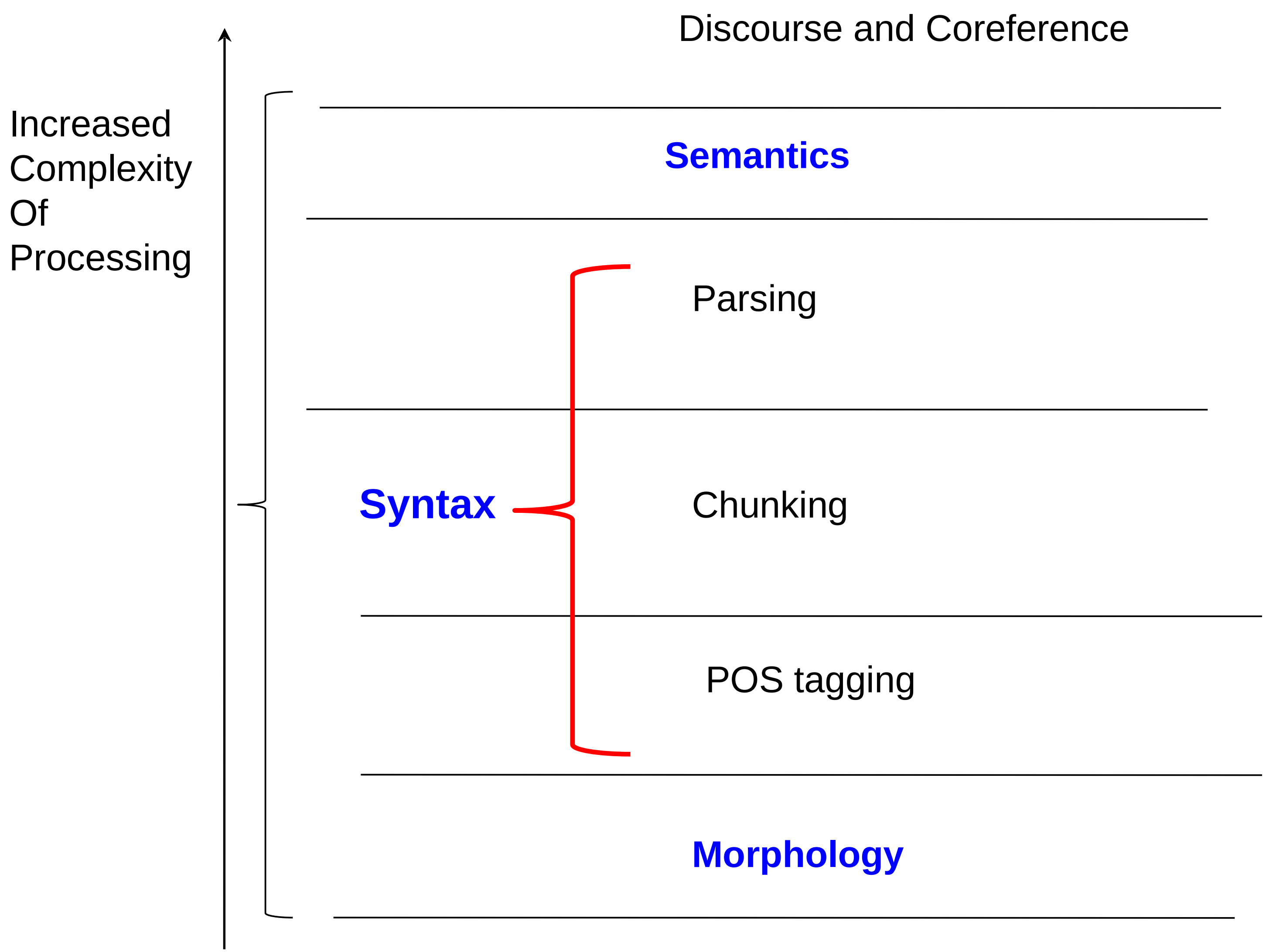
chatGPT response

- Buffalo (noun): Refers to the city of Buffalo, which is in the state of New York in the United States.
- buffaloes (verb): Means to bully, confuse, or baffle.
- Buffalo (noun): Refers to the animal, specifically the American bison.

describes a scenario where bison from Buffalo are being intimidated or confused by other bison from the same place, who, in turn, are intimidating or confusing other bison. It's a playful sentence that plays on the multiple meanings of the word "Buffalo" as a place

CORRECT!!

NLP Layers


Discourse and Coreference
Increased
Complexity
Of
Processing
Semantics
Parsing
Syntax
Chunking
POS tagging
Morphology

Tools for Language Computation

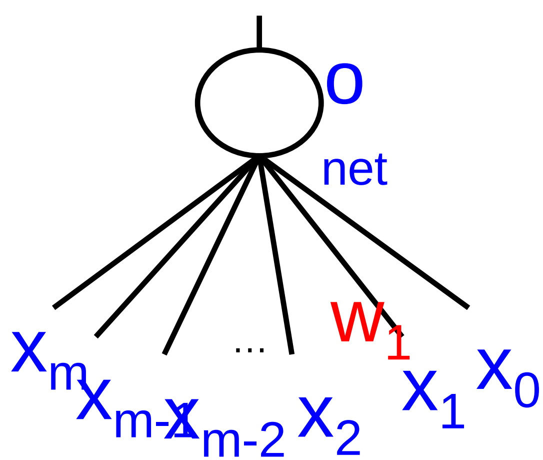
Softmax

$$\sigma(\bar{Z})_i = \frac{e^{Z_i}}{\sum_{j=1}^{K} e^{Z_j}}$$

Cross Entropy

$$H(P,Q) = -\sum_{x=1,N}\sum_{k=1,C} P(x,k)\log_2 Q(x,k)$$

BP



$$\frac{L}{w_1} = \frac{L}{o}\cdot\frac{o}{net}\cdot\frac{net}{w_1}$$

$$L = -t\log o - (1-t)\log(1-o)$$

$$\Delta w_1 = -\frac{L}{w_1} = (t-o)x_1$$

Perspective-3: Language Modelling

What does a Model do? **PREDICT**

$$s = ut + \frac{1}{2}at^2$$

From <*u*, *a*, *t*> compute (predict!) *s*

What does a Language Model do?

PREDICT NEXT LANGUAGE OBJECT

- QA*:*
  - *What is the capital of India →Delhi*

- Summary:
  - *Delhi is a large city. The city has large number of people residing and passing through. Large businesses are transacted here. The cultural activities are numerous. →Delhi is a large, populous and busy city*

What does a Language Model do?

PREDICT PROPERTY OF LANGUAGE OBJECTS

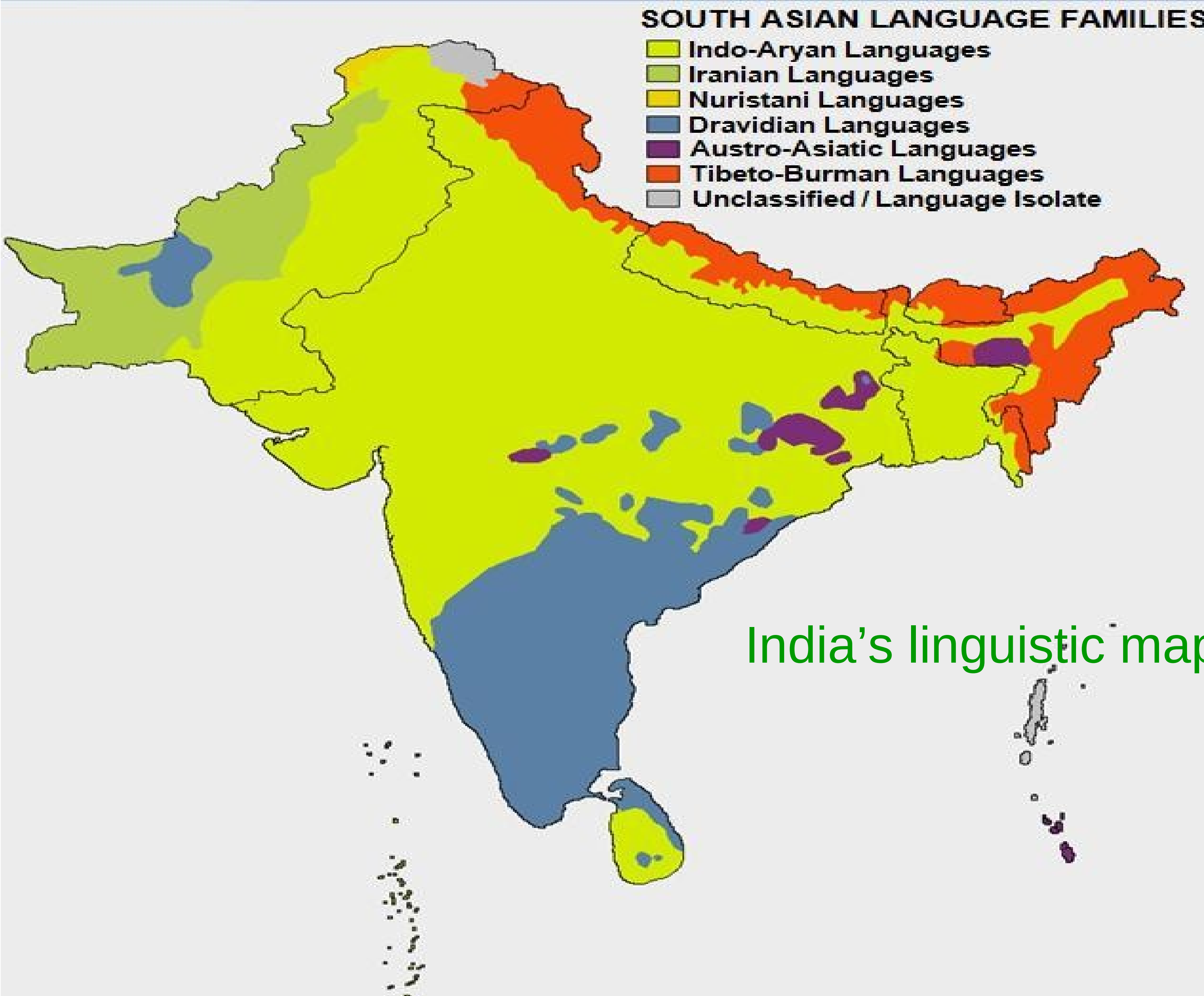
- Classification/regression: sentiment analysis:
  - *The movie was wonderful* → *+ve sentiment*

LLMs too Predict, but very long Context

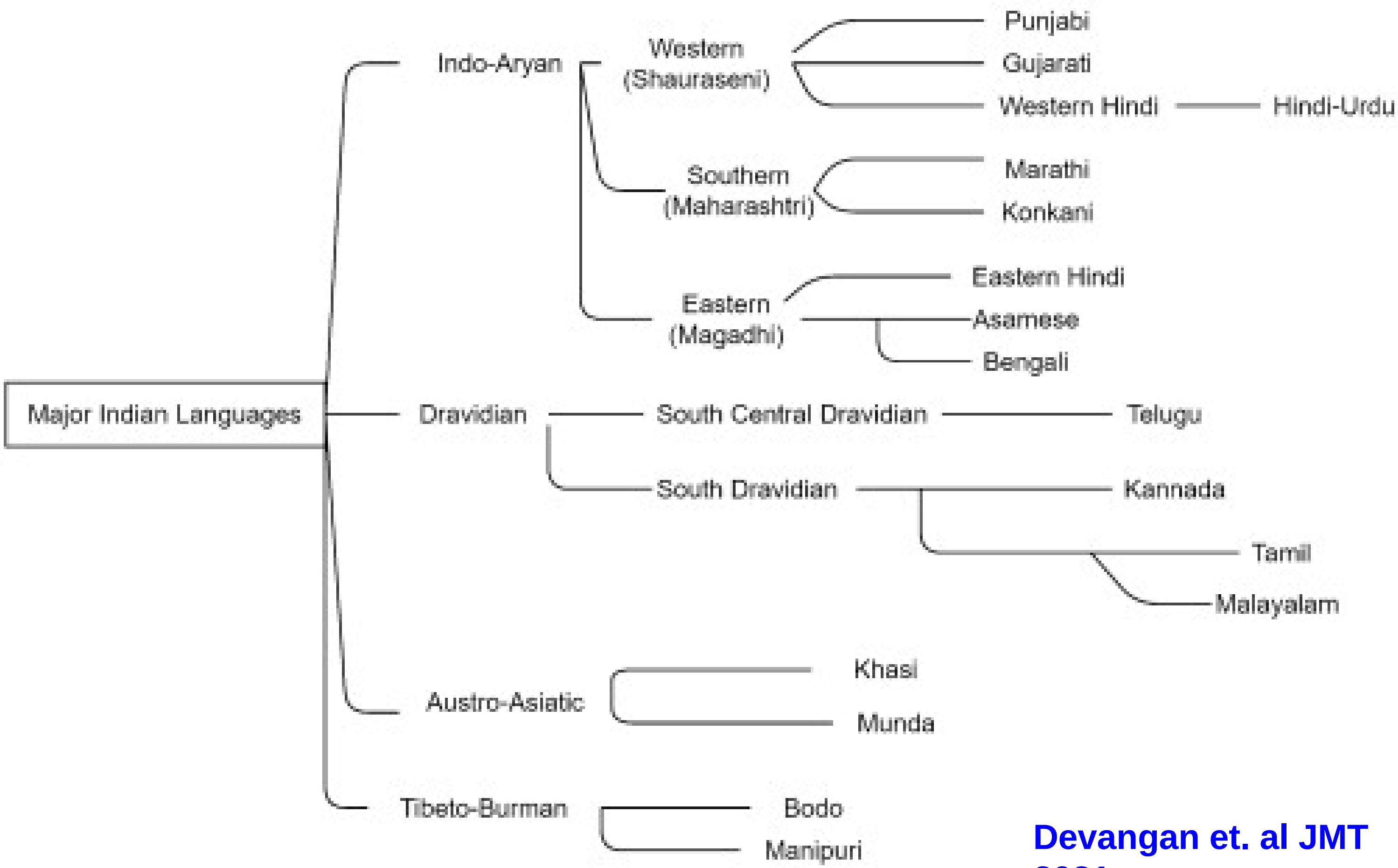
- 1 trillion (openAI) to 4 trillion (Amazon's Olympus) parameters

- Interesting: Olympus expenditure- $4billion, about 33K Cr INR; Goa budget about 25K Cr INR

Multilinguality



India’s linguistic map



Devangan et. al JMT 2021

Fig. 1 Tree diagram to illustrate the language closeness of major Indian languages

NLP of Rare Language Phenomena

Sarcasm

Long Line of Work in Rare Language Phenomena

- **Sarcasm** (ACL 20, WASSA 19, ACM CSUR 17, ACL 17, EMNLP 16, AAAI 16, ACL 15)

- **Humour** Detection and Generation (EMNLP 2021, AACL-IJCNLP 20)

- **Metaphor and Hyperbole** (ACL Findings 23)

- All these phenomena are grounded in the linguistic and cognitive phenomenon of **INCONGRUITY**

Sarcasm

Sarcasm: Etymology

- Greek: ‘*sarkasmós*’: ‘to tear flesh with teeth’

- Sanskrit: ‘*vakrokti*’: ‘a twisted (vakra) utterance (ukti)’

Multimodality is important

Pax: *thank you for sending me to Delhi and my luggage to Mumbai ! Brilliant service!!!*

Chatbot: *Thanks for the appreciation* 🙏

sarcasm

Impact of Sarcasm on Sentiment Analysis

	Precision (Sarc)	Precision (Non-sarc)
Conversation Transcripts		
MeaningCloud[1]	20.14	49.41
NLTK (Bird, 2006)	38.86	81
Tweets		
MeaningCloud[1]	17.58	50.13
NLTK (Bird, 2006)	35.17	69

[1] www.meaningcloud.com

Incongruity: at the heart of things!

- *I love being ignored*
- *3:00 am work YAY. YAY.*
- *Up all night coughing. yeah me!*
- *No power, Yes! Yes! Thank you storm!*
- *This phone has an awesome battery back-up of 2 hour (Sarcastic)*

Two kinds of incongruity

- **Explicit incongruity**
  - Overtly expressed through sentiment words of both polarities
  - Contribute to almost 11% of sarcasm instances

    'I <u>love</u> being <u>ignored</u>'
- **Implicit incongruity**
  - Covertly expressed through phrases of implied sentiment

    'I <u>love</u> this paper so much that I <u>made a doggy bag out of</u> it'

Sarcasm and Sense Ambiguity

Oh! Its so nice of you to give me a ring early in the morning!

Good to see you help dog bite victim!

Sarcasm Detection Using Semantic Incongruity

Aditya Joshi, Vaibhav Tripathi, Kevin Patel, Pushpak Bhattacharyya and Mark Carman, *Are Word Embedding-based Features Useful for Sarcasm Detection?*, **EMNLP 2016**, Austin, Texas, USA, November 1-5, 2016.

Also covered in: How Vector Space Mathematics Helps Machines Spot Sarcasm, MIT Technology Review, 13th October, 2016.

www.cfilt.iitb.ac.in/sarcasmsuite/

Feature Set

Lexical	
Unigrams	Unigrams in the training corpus
Pragmatic	
Capitalization	Numeric feature indicating presence of capital letters
Emoticons & laughter expressions	Numeric feature indicating presence of emoticons and 'lol's
Punctuation marks	Numeric feature indicating presence of punctuation marks
Implicit Incongruity	
Implicit Sentiment Phrases	Boolean feature indicating phrases extracted from the implicit phrase extraction step
Explicit Incongruity	
#Explicit incongruity	Number of times a word is followed by a word of opposite polarity
Largest positive /negative subsequence	Length of largest series of words with polarity unchanged
#Positive words	Number of positive words
#Negative words	Number of negative words
Lexical Polarity	Polarity of a tweet based on words present

Datasets

Name	Text-form	Method of labeling	Statistics
Tweet-A	Tweets	Using sarcasm-based hashtags as labels	5208 total, 4170 sarcastic
Tweet-B	Tweets	Manually labeled (Given by Riloff et al(2013))	2278 total, 506 sarcastic
Discussion-A	Discussion forum posts (IAC Corpus)	Manually labeled (Given by Walker et al (2012))	1502 total, 752 sarcastic

Results

Features	P	R	F
Original Algorithm by Riloff et al. (2013)			
Ordered	0.774	0.098	0.173
Unordered	0.799	0.337	0.474
Our system			
Lexical (**Baseline**)	0.820	0.867	0.842
Lexical+Implicit	0.822	0.887	0.853
Lexical+Explicit	0.807	0.985	0.8871
All features	0.814	0.976	**0.8876**

Approach	P	R	F
Riloff et al. (2013) (**best reported**)	0.62	0.44	0.51
Maynard and Greenwood (2014)	0.46	0.38	0.41
Our system (all features)	**0.77**	**0.51**	**0.61**

Tweet-B

Tweet-A

Features	P	R	F
Lexical (**Baseline**)	0.645	0.508	0.568
Lexical+Explicit	0.698	0.391	0.488
Lexical+Implicit	0.513	0.762	0.581
All features	0.489	0.924	**0.640**

Discussion-A

Capturing Incongruity Using Word Vectors

Use **similarity** of word embeddings

”A man needs a woman like a fish needs bicycle”

Word2Vec *similarity(man,woman)*= 0.766
Word2Vec *similarity(fish, bicycle)*= 0.131

Results (1/2)

Features	P	R	F
	Baseline		
Unigrams	67.2	78.8	72.53
S	64.6	75.2	69.49
WS	67.6	51.2	58.26
Both	67	52.8	59.05

	LSA			GloVe			Dependency Weights			Word2Vec		
	P	R	F	P	R	F	P	R	F	P	R	F
L	73	79	75.8	73	79	75.8	73	79	75.8	73	79	75.8
+S	81.8	78.2	**79.95**	81.8	79.2	**80.47**	81.8	78.8	80.27	80.4	80	**80.2**
+WS	76.2	79.8	77.9	76.2	79.6	77.86	81.4	80.8	81.09	80.8	78.6	79.68
+S+WS	77.6	79.8	78.68	74	79.4	76.60	82	80.4	**81.19**	81.6	78.2	79.86
G	84.8	73.8	78.91	84.8	73.8	78.91	84.8	73.8	**78.91**	84.8	73.8	**78.91**
+S	84.2	74.4	**79**	84	72.6	77.8	84.4	72	77.7	84	72.8	78
+WS	84.4	73.6	78.63	84	75.2	**79.35**	84.4	72.6	78.05	83.8	70.2	76.4
+S+WS	84.2	73.6	78.54	84	74	78.68	84.2	72.2	77.73	84	72.8	78
B	81.6	72.2	76.61	81.6	72.2	76.61	81.6	72.2	76.61	81.6	72.2	76.61
+S	78.2	75.6	**76.87**	80.4	76.2	**78.24**	81.2	74.6	**77.76**	81.4	72.6	76.74
+WS	75.8	77.2	76.49	76.6	77	76.79	76.2	76.4	76.29	81.6	73.4	77.28
+S+WS	74.8	77.4	76.07	76.2	78.2	77.18	75.6	78.8	77.16	81	75.4	**78.09**
J	85.2	74.4	79.43	85.2	74.4	79.43	85.2	74.4	79.43	85.2	74.4	79.43
+S	84.8	73.8	78.91	85.6	74.8	79.83	85.4	74.4	79.52	85.4	74.6	**79.63**
+WS	85.6	75.2	**80.06**	85.4	72.6	78.48	85.4	73.4	78.94	85.6	73.4	79.03
+S+WS	84.8	73.6	78.8	85.8	75.4	**80.26**	85.6	74.4	**79.6**	85.2	73.2	78.74

Table 3: Performance obtained on augmenting word embedding features to features from four prior works, for four word embeddings; L: Liebrecht et al. (2013), G: González-Ibánez et al. (2011a), B: Buschmeier et al. (2014) , J: Joshi et al. (2015)

Numerical Sarcasm

Illustrates *need* for

Rule Based → Classical ML → Deep Learning

Abhijeet Dubey, Lakshya Kumar, Arpan Somani, Aditya Joshi and Pushpak Bhattacharyya, *"When Numbers Matter!": Detecting Sarcasm in Numerical Portions of Text*, 10th Workshop on Computational Approaches to Subjectivity, Sentiment & Social Media Analysis (**WASSA 2019**), Minneapolis, USA, 7 June, 2019.

About 17% of sarcastic tweets have origin in number

1- This phone has an awesome battery back-up of 38 hours (Non-sarcastic)

2- This phone has a terrible battery back-up of 2 hours (Non-sarcastic)

3- This phone has an awesome battery back-up of 2 hour (Sarcastic)

Interesting question: why people use sarcasm?

- Dramatization, Forceful Articulation, lowering defence and then attack!

Comparison of results (1: sarcastic, 0: non-sarcastic)

Approaches	Precision			Recall			F-score		
	P(1)	P(0)	P(avg)	R(1)	R(0)	R(avg)	F(1)	F(0)	F(avg)
Past Approaches									
Buschmeier et.al.	0.19	0.98	0.84	0.99	0.07	0.24	0.32	0.13	0.16
Liebrecht et.al.	0.19	1.00	0.85	1.00	0.07	0.24	0.32	0.13	0.17
Gonzalez et.al.	0.19	0.96	0.83	0.99	0.06	0.23	0.32	0.12	0.15
Joshi et.al.	0.20	1.00	0.86	1.00	0.13	0.29	0.33	0.23	**0.25**
Rule-Based Approaches									
Approach-1	0.53	0.87	0.81	0.39	0.92	0.83	0.45	0.90	**0.82**
Approach-2	0.44	0.85	0.78	0.28	0.92	0.81	0.34	0.89	0.79
Machine-Learning Based Approaches									
SVM	0.50	0.95	0.87	0.80	0.82	0.82	0.61	0.88	**0.83**
KNN	0.36	0.94	0.84	0.81	0.68	0.70	0.50	0.79	0.74
Random Forest	0.47	0.93	0.85	0.74	0.81	0.80	0.57	0.87	0.82
Deep-Learning Based Approaches									
CNN-FF	**0.88**	**0.94**	**0.93**	**0.71**	**0.98**	**0.93**	**0.79**	**0.96**	**0.93**
CNN-LSTM-FF	0.82	0.94	0.92	0.72	0.96	0.92	0.77	0.95	0.92
LSTM-FF	0.76	0.93	0.90	0.68	0.95	0.90	0.72	0.94	0.90

back

Humble Bragging

Opposite of Sarcasm: can fool a sarcasm classifier

“My life is miserable, have to sign 500 autographs everyday”: Exposing Humblebragging, the Brags in Disguise

Sharath H N, Saprativa Bhattacharjee, Biplab Banerjee and Pushpak Bhattacharyya. My life is miserable, have to sign 500 autographs everyday: Exposing Humblebragging, the Brags in Disguise. ACL Findings-2025. Vienna, Austria, July 27-August 1, 2025

Problem Statement

Task: Classification

Input: Text

Output: 1/0: humble bragging or not

Example:

- *Can someone tell the awards committee to chill? Running out of shelf space here!* 🤷‍♂️ *- Humblebragging*
- *Oh, I love last-minute changes. It's my favorite way to live! - Not Humblebragging*

Methodology

Definition as a part of input to LLM

<definition>: A humble brag comprises the following Components:

1. Brag: The segment of the text that explicitly conveys the act of bragging

2. Brag Theme: The overarching theme or specific category of the brag embedded within the statement

3.Possible categories: Looks and Attractiveness, Achievements, Performance at Work, Money and Wealth, Intelligence, Personality, Social Life, Miscellaneous

Definition as a part of input to LLM (Part 2)

3. Humble Mask:
   - The element of the text that adopts a modest or self-deprecating tone to obscure or mitigate the act of bragging

4. Mask Type:
   - Specifies whether the humble mask adopts a modest tone or a self-deprecating approach

Now you are about to classify if a given sentence is a humble brag or not using the above definition

Input-Output as Question Answering

Input to LLM:

<definition>: **B, BT, M, MT**

B: *Can someone tell the awards committee to chill*

BT: Achievements

M: R*unning out of shelf space here!*

MT: *Complain*

<question>: Is the given text humblebragging or not? Answer in Yes or No only

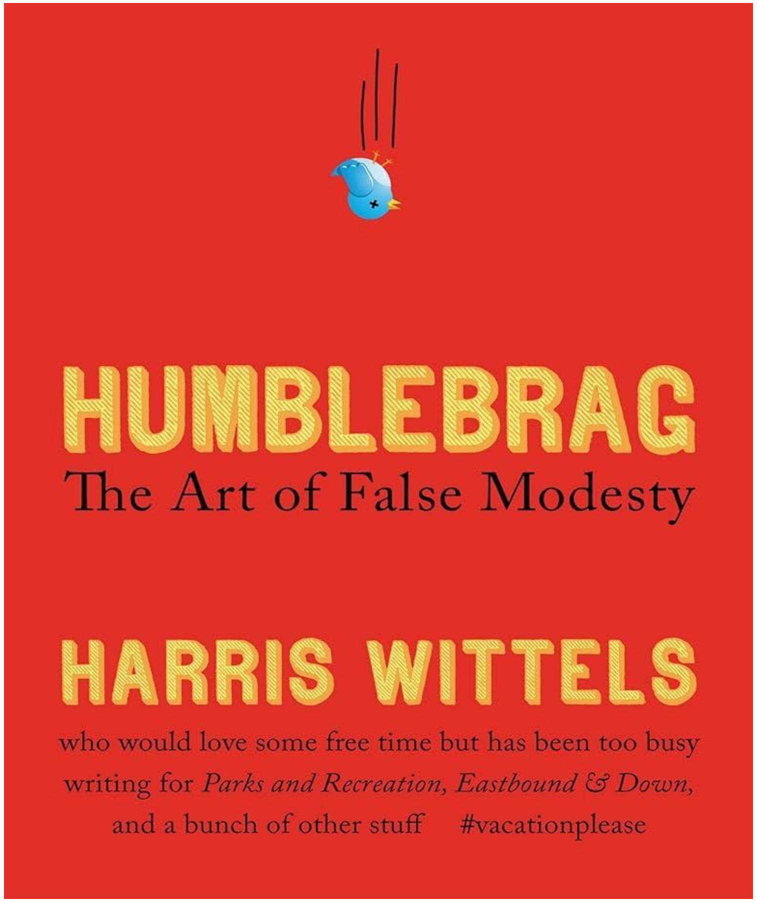
<x>”*Can someone tell the awards committee to chill? Running out of shelf space here!* 🤷‍♂️“

Output from LLM:

<answer>: Yes

Foundations

What is a Humblebrag?



“a specific type of brag that masks the boasting part of a statement in a faux-humble guise. The false humility allows the offender to boast about their "achievements" without any sense of shame or guilt”: Harris Wittels

Computational Humble Bragging: a novel task

- **Distinct from Sarcasm:** While both exhibit incongruity, humblebragging serves a different social purpose like self-promotion masked as modesty, not ridicule.
  - HumbleBragging: *Honestly, managing wealth is stressful in its own way. I miss the simplicity of not having to think about diversification.*
  - Sarcasm: *Being broke is so freeing. I never have to worry about where to invest!*
- **Underexplored in NLP:** prevalence in online discourse, but no public benchmarks or standard datasets.
- **Potential Impact**:
  - Intent-aware sentiment analysis
  - Personality profiling and social behavior modeling
  - Enables socially intelligent, human-like chatbots

Difficulty in Identifying Humblebrag

High context-dependence

- **“It was so foolish of me to spill wine on my $2000 suit" may signal self-deprecation or a subtle brag about owning an expensive suit but can also be a genuine concern.**

Cultural awareness

- **“Told Chris the lighting made me look bad and he just rolled his eyes….classic Marvel sets.” can only be understood by people familiar with the Marvel cast.**

Ambiguity in literal statement.

- **“Too tired of all these interviews”**
  - **Job interviews: Genuine concern/frustration**
  - **Celebrity Interviews: Subtle humblebragging**

Humblebragging in Psychology

Sezer, Gino, and Norton (2018): first systematic analysis of humblebragging from a psychological perspective:

- **Definition**: Humblebragging is a form of self-promotion masked by modesty or complaint (e.g., “I hate how all my clothes are too big now”).

- **Key Findings**:
  - **More detestable** than both direct bragging and genuine complaining
  - Perceived as **insincere** and manipulative.
  - Backfires because it violates expectations of authenticity

- **Implication**: Although intended to boost social perception, humblebragging often **reduces trust and likability**—making it a flawed self-presentation strategy.

Automatic Humblebragging Detection: our contribution

- Task Formalization as binary classification
- Novel 4-tuple definition capturing *brag, brag theme, humble mask, and mask type*
- A new dataset (HB24): 3,340 synthetic humblebrags and 5,431 human written non-humblebrags
- Benchmarking of encoders (BERT, RoBERTa), and decoder LLMs (GPT-4o, LLaMA, etc.) across zero-shot, definition-augmented, and fine-tuned settings
- Demonstration of our best model (GPT-4o) that achieved an F1-score of 0.88, surpassing average human annotator performance

4-Tuple Definition of Humblebragging

To systematically detect humblebragging, we introduce a formal representation comprising a four-tuple definition that captures its key linguistic and contextual properties.

HB = ⟨B, BT, HM, MT⟩

Component	Description
B	**Brag** – The explicit bragging content
BT	**Brag Theme** – Category of the brag (e.g., performance, appearance)
HM	**Humble Mask** – Modest or self-deprecating language masking the brag
MT	**Mask Type** – Type of mask: *Modest* or *Complaint*

Example: Complaint based humblebrags

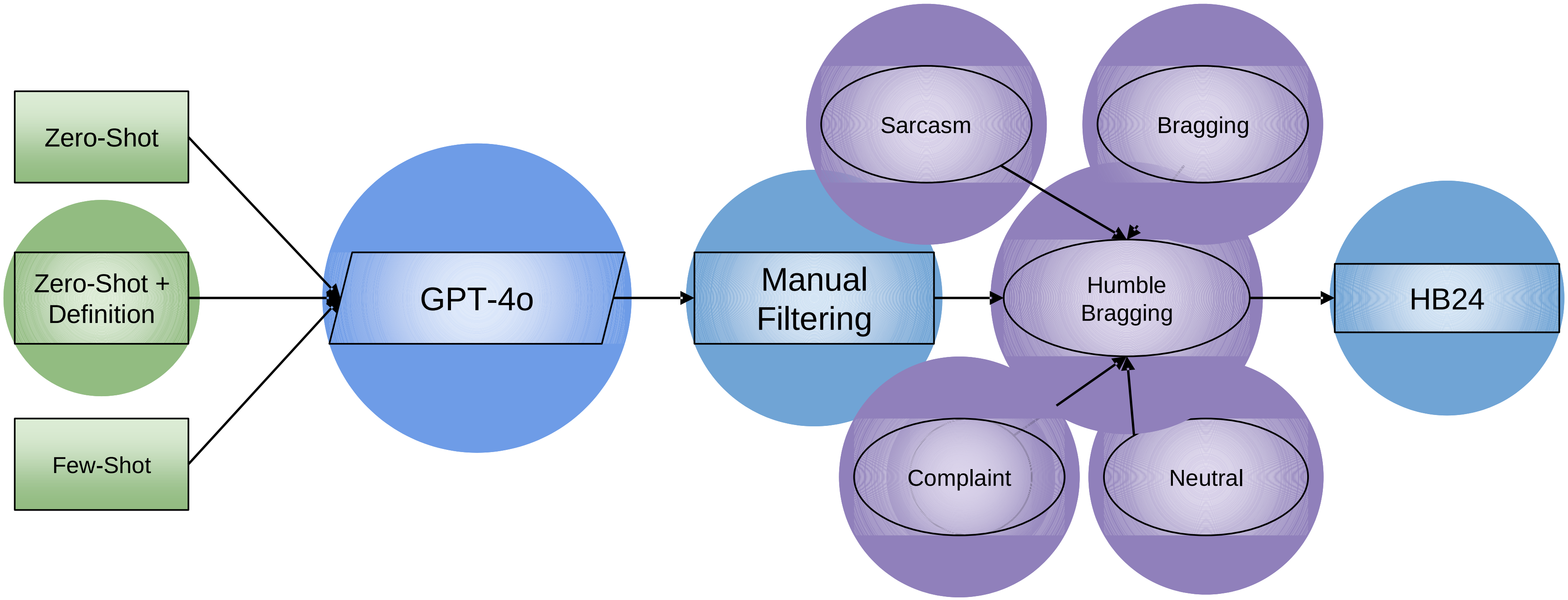
"Ugh, I can’t believe I got promoted to lead the entire team. So stressful!"

B: "I can’t believe I got promoted to lead the entire team.";
BT : Performance at work
HM : "Ugh," and "So stressful!"
MT : Complaint.

“Why does my passport get filled so quickly!!! It looks like I need an assistant to manage my passports”

B: "*my passport gets filled so quickly*";
BT : Travel/Social Life
HM : "need an assistant to manage"
M T : Complaint.

Dataset- HB24: A mixture of Human written and machine generated texts



Results

Model	Precision	Recall	F1-Score	Accuracy
Baseline	0.25	0.50	0.34	0.51
Human 1	0.89	0.81	0.85	0.86
Human 2	0.86	0.81	0.84	0.84
Human 3	0.82	0.51	0.63	0.70
Average	0.86	0.71	0.77	0.80
Logistic Regression	0.68	0.58	0.53	0.59
SVM	0.72	0.61	0.56	0.62
BERT-Large-Uncased (F)	0.76	0.50	0.61	0.68
RoBERTa-Large (F)	**0.91**	0.62	0.74	0.78
GPT-4o (Z)	0.78	0.94	0.85	0.84
GPT-4o (Z+D)	**0.91**	0.85	**0.88**	**0.89**
GPT-3.5 (Z)	0.65	0.60	0.57	0.61
GPT-3.5 (Z+D)	0.76	0.75	0.75	0.75
Qwen2.5-7B-Instruct (Z)	0.82	0.35	0.49	0.64
Qwen2.5-7B-Instruct (Z+D)	0.85	0.50	0.63	0.71
Qwen2.5-7B-Instruct (F)	0.85	0.40	0.54	0.67
Mistral-7B-Instruct-v0.3 (Z)	0.55	0.96	0.70	0.60
Mistral-7B-Instruct-v0.3 (Z+D)	0.55	0.96	0.70	0.60
Llama-3.1-8B-Instruct (Z)	0.49	**0.99**	0.66	0.49
Llama-3.1-8B-Instruct (Z+D)	0.62	0.88	0.72	0.68
Llama-3.1-8B-Instruct (F)	0.87	0.72	0.79	0.81
Gemma-1.1-7b-it (Z)	0.57	0.57	0.57	0.57
Gemma-1.1-7b-it (Z+D)	0.53	0.83	0.65	0.56
Gemma-1.1-7b-it (F)	0.71	0.44	0.60	0.71
Vicuna-7b-v1.5 (Z)	0.60	0.28	0.38	0.55
Vicuna-7b-v1.5 (Z+D)	0.62	0.51	0.56	0.61

Zero shot +
Definition

Analysis

- **LLMs with Definitions Excelled**: Providing decoder models with our 4-tuple definition (Z+D) consistently improved F1-scores across all models, validating the utility of a structured definition.

- **GPT-4o Outperformed Humans**: GPT-4o achieved the highest F1-score of 0.88, even surpassing the best human annotator (F1-score 0.85), showcasing LLMs' potential in nuanced language understanding.

- **Fine-Tuning Helped Most Models**: Fine-tuning with the HB24 dataset improved performance for both encoder and decoder models, especially RoBERTa and LLaMA.

- **Task is Non-Trivial**: Variance in human annotator performance and frequent model-human disagreements indicate that humblebrag detection is inherently challenging and subjective.

A Multi-task Framework for Hyperbole and Metaphor Detection

Naveen Badathala, Abisek Rajakumar Kalarani, Tejpalsingh Siledar and Pushpak Bhattacharyya, A Match Made in Heaven: A Multi-task Framework for Hyperbole and Metaphor Detection, ACL 2023 Findings, Toronto, July 9-14, 2023.

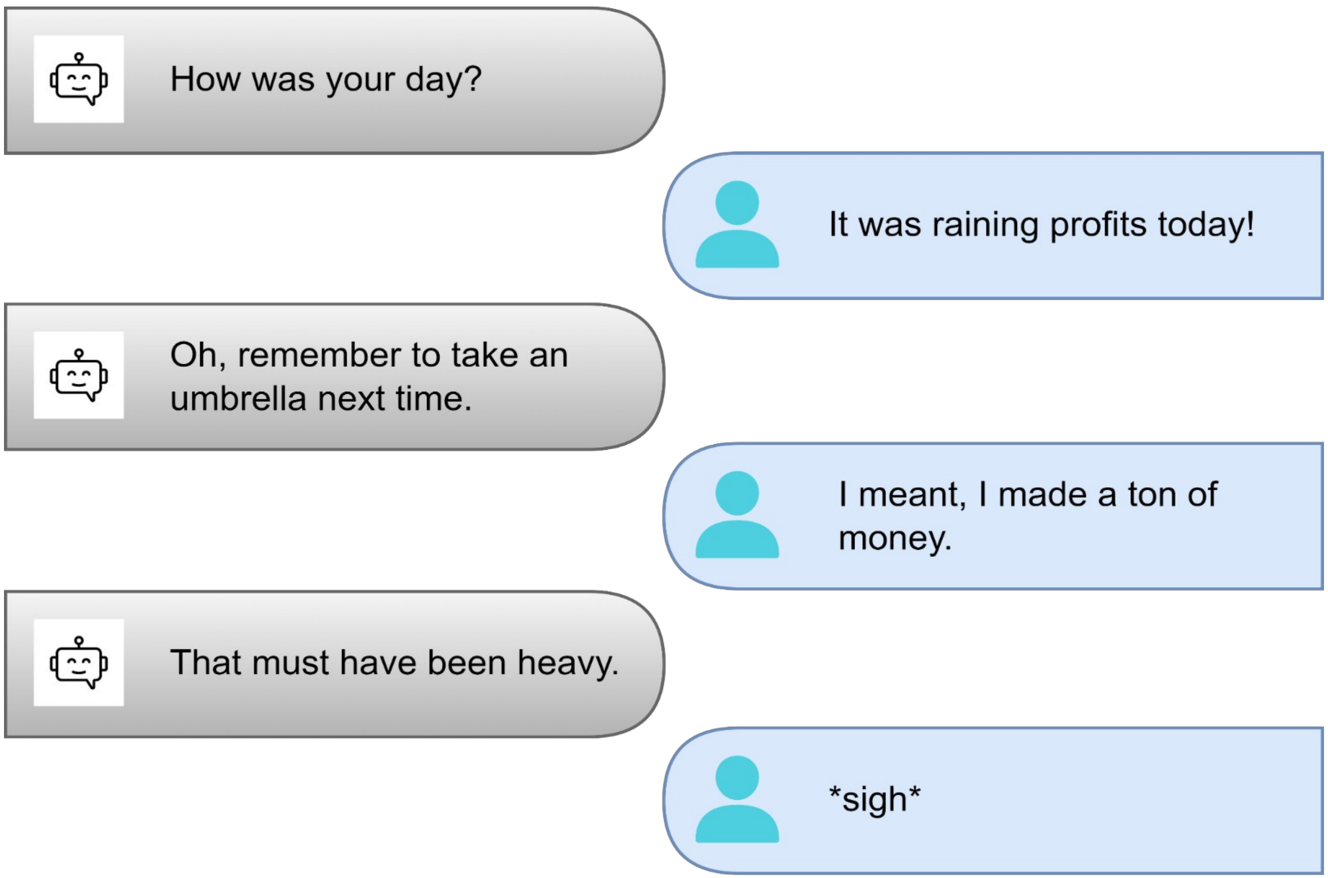
Hyperbole and Metaphor

- A hyperbole is a figurative language in which the literal meaning is exaggerated intentionally
- Metaphor makes implicit comparisons to something that is literally not true

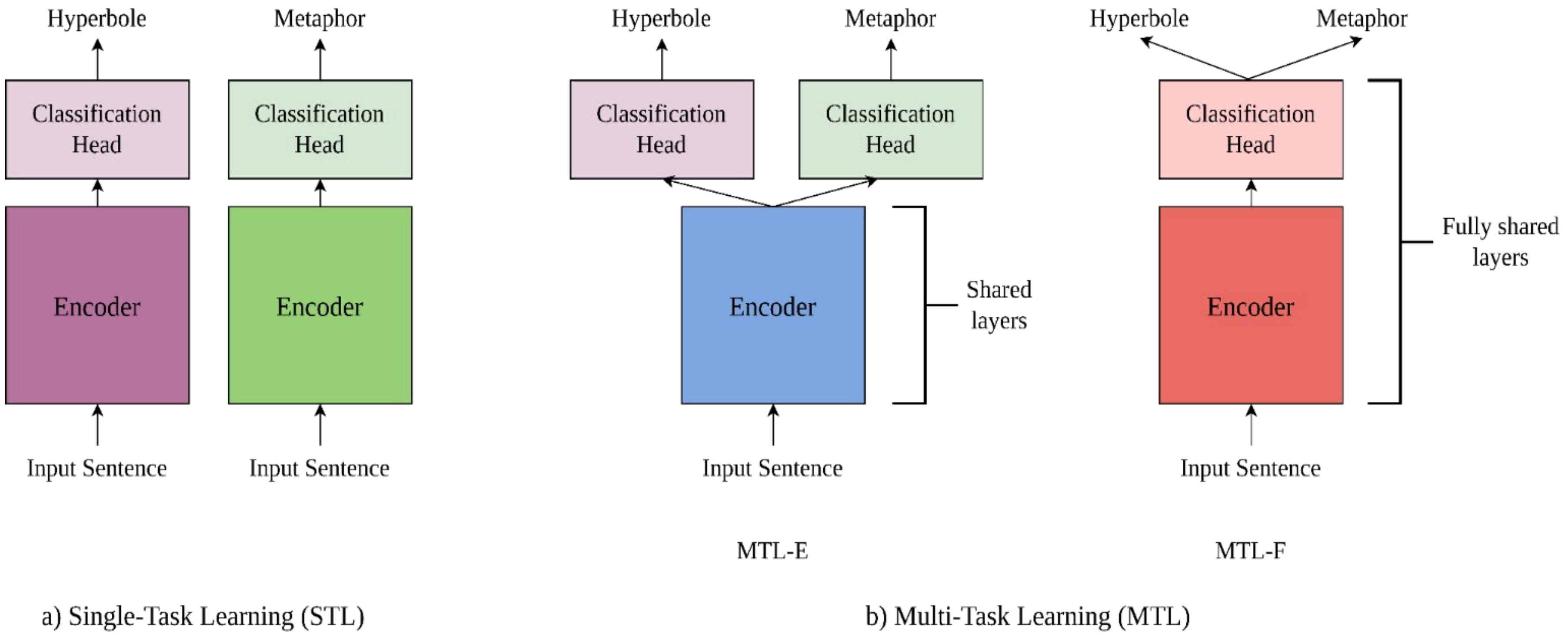
Input	**Hyperbole (0/1)**	**Metaphor (0/1)**
The principal is angry	0	0
I will walk a thousand miles to meet you.	1	0
Life is a journey	0	1
The principal is going to cook us	1	1

Motivation (1/2)

Hyperbole and metaphor are common and their detection is important, e.g., in the following with a chatbot



Schematic



Input: Sentence

Output: Hyperbole or Metaphor

Dataset

- HYPO, HYPO-L (Hyperbole Datasets)
- LCC, TroFi (Metaphor Datasets)

Dataset (# sentences)	Hyp.	Met.	# sent.
HYPO (1,418)	✓	✓	515
	✓	✗	194
	✗	✓	107
	✗	✗	602
HYPO-L (3,326)	✓	✓	237
	✓	✗	770
	✗	✓	19
	✗	✗	2,200

Table 1: Statistics of annotated hyperbole datasets with metaphor labels, where Hyp. means hyperbole, Met. means metaphor, and #sent is the number of sentences.

Dataset (# sentences)	Met.	Hyp.	# sent.
TroFi (3,838)	✓	✓	209
	✓	✗	1,710
	✗	✓	235
	✗	✗	1,684
LCC (7,542)	✓	✓	615
	✓	✗	3,187
	✗	✓	144
	✗	✗	3,596

Table 2: Statistics of annotated metaphor datasets with hyperbole labels, where Hyp. means hyperbole, Met. means metaphor, and #sent is the number of sentences.

STL vs. MTL Results - MTL superior (1/2)

Task	Model	Hyperbole			Metaphor		
		Precision	Recall	F1	Precision	Recall	F1
STL	$BERT_{large}$	0.827	0.801	0.811	0.751	0.686	0.711
	$ALBERT_{large}$	0.845	0.871	0.856	0.695	0.736	0.713
	$RoBERTa_{large}$	**0.883**	0.848	0.864	0.801	0.709	0.745
MTL	$BERT_{large}$	0.853	0.824	0.836	0.799	0.686	0.729
	$ALBERT_{large}$	0.847	0.878	0.860	0.757	**0.761**	0.753
	$RoBERTa_{large}$	0.879	**0.884**	**0.881**	**0.826**	0.752	**0.787**

10-fold cross validation over three different runs for hyperbole and metaphor detection task on the **HYPO** dataset.

Task	Model	Hyperbole			Metaphor		
		Precision	Recall	F1	Precision	Recall	F1
STL	$BERT_{large}$	0.670	0.598	0.630	0.561	0.466	0.506
	$ALBERT_{large}$	0.649	0.542	0.589	0.513	0.414	0.456
	$RoBERTa_{large}$	0.688	0.651	0.667	0.591	0.543	0.563
MTL	$BERT_{large}$	0.655	0.619	0.638	0.552	0.464	0.503
	$ALBERT_{large}$	0.638	0.593	0.614	0.498	0.385	0.430
	$RoBERTa_{large}$	**0.706**	**0.668**	**0.687**	**0.599**	**0.554**	**0.572**

10-fold cross validation over three different runs for hyperbole and metaphor detection task on the **HYPO-L** dataset.

STL vs. MTL Results - MTL superior (2/2)

Task	Model	Hyperbole			Metaphor		
		Precision	Recall	F1	Precision	Recall	F1
STL	$BERT_{lg}$	0.557	0.412	0.466	0.531	0.559	0.538
	$ALBERT_{xxl2}$	0.424	0.234	0.294	0.489	0.430	0.454
	$RoBERTa_{lg}$	**0.607**	0.446	0.496	0.542	0.469	0.490
MTL-F	$BERT_{lg}$	0.565	0.433	0.486	0.556	0.525	0.540
	$ALBERT_{xxl2}$	0.487	0.241	0.312	0.516	0.457	0.475
	$RoBERTa_{lg}$	0.605	**0.529**	**0.561**	**0.565**	**0.587**	**0.573***

10-fold cross validation over three different runs for hyperbole and metaphor detection task on the label balanced **TroFi** dataset.

Task	Model	Hyperbole			Metaphor		
		Precision	Recall	F1	Precision	Recall	F1
STL	$BERT_{large}$	0.649	0.542	0.589	0.758	0.736	0.745
	$ALBERT_{large}$	0.591	0.546	0.564	0.723	0.757	0.739
	$RoBERTa_{large}$	**0.692**	0.604	0.645	**0.802**	0.787	0.794
MTL	$BERT_{large}$	0.633	0.531	0.575	0.750	0.774	0.760
	$ALBERT_{large}$	0.614	0.425	0.499	0.709	0.785	0.744
	$RoBERTa_{large}$	0.630	**0.691**	**0.659**	0.798	**0.812**	**0.805**

10-fold cross validation over three different runs for hyperbole and metaphor detection task on the label balanced **LCC** dataset.

Hyperbole Detection Results - New SOTA!

	Model	P	R	F1
Baselines	LR+QQ	0.679	0.745	0.710
	NB+QQ	0.689	0.696	0.693
	$BERT_{base}$	0.711	0.735	0.709
	$BERT_{base}$+QQ	0.650	0.765	0.671
	$BERT_{base}$+PI	0.754	0.814	0.781
Ours	$RoBERTa_{lg}$ STL	**0.883**	0.848	0.864
	$RoBERTa_{lg}$ MTL-E	0.859	0.878	0.867
	$RoBERTa_{lg}$ MTL-F	0.879	**0.884**	**0.881**

Precision (P), recall (R) and F1 score for baseline models compared to our work on HYPO dataset

Qualitative Analysis

Sentences	Actual	MTL-F	STL	
			HD	MD
Your plan is too risky, it's a suicide.	H, M	H, M	NH	NM
I'm not staying here any longer!	NH, NM	NH, NM	H	NM
This kind of anger rages like a sea in a storm.	H, NM	H, NM	H	M
My ex boyfriend! Treacherous person!	NH, NM	NH, NM	H	M
They cooked a turkey the size of a cow.	H, M	H, M	H	NM
Her strength awoke in poets an abiding love.	NH, M	NH, M	H	M
My sister is a vortex of intelligence in space.	H, M	H, M	H	M
The act of love strongly resembles severe pain.	NH, NM	NH, NM	NH	NM

Cases where the MTL-F performs better than the STL. H-Hyperbole, NH-Non Hyperbole, M-Metaphor, NM-Non Metaphor. Red indicates incorrect detection.

- MTL consistently outperforms STL
- We investigate all four combinations of hyperbole and metaphor cases and observe a similar pattern

NLP in Low Resource Settings

Part-of-speech Tagging for Extremely Low-resource Indian Languages

Sanjeev Kumar, Preethi Jyothi, Pushpak Bhattacharyya
Computer Science & Engineering, IIT Bombay, India, ACL 2024 Findings

Paper, code, and data available at: https://github.com/snjev310/acl-24-pos

Introduction

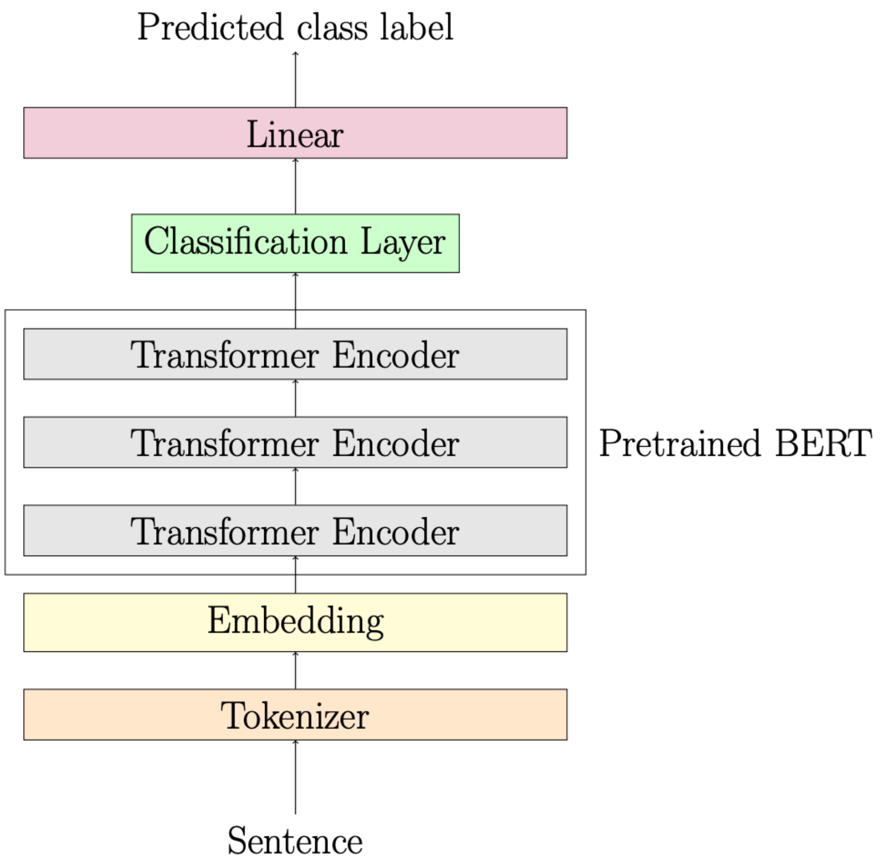
- Modern NLP systems require large amounts of data.
- Extremely low-resource languages (ELRL) have huge data sparsity, and are less documented.
- Even for **shallow NLP tasks,** ELRL do not have datasets.
- Development of NLP tools will **help preserve multilinguality.**
- India is a multilingual country with more than 1369 languages and five main language families.
- Our primary objective is to develop a POS tagger for three extremely low-resourced languages: Angika, Magahi, and Bhojpuri.

Task: POS Tagger for ELRLs: Angika, Bhojpuri, Magahi

- Bihari language group
- Belong to the Indo-Aryan language family
- Devanagari script
- Subject-Object-Verb (SOV)

Models

- MuRIL, RemBERT, and XLM-R-Large
- Added a classification layer on top of the encoder output
- None of these models are trained on Angika, Magahi, and Bhojpuri



Experiments and Results

- Zero-shot setting: Train the model using Hindi POS data, and evaluate for POS-tagging accuracies on Angika, Magahi, and Bhojpuri evaluation sets

		Hindi			Angika			Magahi			Bhojpuri		
		P	**R**	**F1**	**P**	**R**	**F1**	**P**	**R**	**F1**	**P**	**R**	**F1**
Hindi FT	MuRIL	0.93	0.94	0.93	0.66	0.72	0.69	0.74	0.78	0.76	0.78	0.80	0.79
	XLM-R large	0.93	0.93	0.93	0.68	0.69	0.69	0.73	0.74	0.74	0.76	0.77	0.76
	RemBERT	0.93	0.94	0.94	0.67	0.71	0.69	0.75	0.77	0.76	0.76	0.77	0.76
	AVG	0.93	0.94	0.93	0.67	0.71	0.69	0.74	0.76	0.75	0.76	0.77	0.77

Simple sub-word strategies (to solve data sparsity problem)

- **Look-back (LB):** Consider the initial token of the word; update the predictions for all other sub-word tokens with the label predicted for the first sub-word token.
- **Look-back-with-score (LBS):** Consider the token logit scores; find the sub-word token with the maximum logit score and retrieve its corresponding POS label.

Simple sub-word strategies results

		Hindi			Angika			Magahi			Bhojpuri		
		P	**R**	**F1**	**P**	**R**	**F1**	**P**	**R**	**F1**	**P**	**R**	**F1**
Hindi FT	MuRIL	0.93	0.94	0.93	0.66	0.72	0.69	0.74	0.78	0.76	0.78	0.80	0.79
	XLM-R large	0.93	0.93	0.93	0.68	0.69	0.69	0.73	0.74	0.74	0.76	0.77	0.76
	RemBERT	0.93	0.94	0.94	0.67	0.71	0.69	0.75	0.77	0.76	0.76	0.77	0.76
	AVG	0.93	0.94	0.93	0.67	0.71	0.69	0.74	0.76	0.75	0.76	0.77	0.77
Look-back	MuRIL	**0.95**	0.94	0.94	**0.80**	**0.77**	**0.77**	**0.84**	**0.82**	**0.83**	**0.85**	**0.83**	**0.84**
	XLM-R large	**0.95**	**0.95**	**0.95**	0.78	0.73	0.74	0.81	0.76	0.77	0.80	0.75	0.75
	RemBERT	0.94	0.94	0.94	0.79	0.73	0.74	0.82	0.79	0.80	0.82	0.78	0.79
	AVG	0.95	0.94	0.94	0.79	0.74	0.75	0.82	0.79	0.80	0.83	0.79	0.79
Look-back with score	MuRIL	**0.95**	**0.95**	**0.95**	**0.80**	0.76	**0.77**	0.83	0.81	0.82	**0.85**	0.82	0.83
	XLM-R large	**0.95**	0.94	0.94	0.79	0.74	0.75	0.83	0.80	0.80	0.83	0.80	0.80
	RemBERT	**0.95**	**0.95**	**0.95**	**0.80**	0.75	0.76	0.83	0.81	0.81	0.84	0.81	0.81
	AVG	0.95	0.95	0.95	0.80	0.75	0.76	0.83	0.81	0.81	0.84	0.81	0.81

Table 1: POS tagging Precision (**P**), Recall (**R**), and F1-score (**F1**) of three PLMs evaluated on our parallel dataset using zero-shot, look-back, and look-back-with-score methods. Results range from 0 to 1 and are averaged across five random seeds.

Per PoS Accuracy

	Hindi FT					Look-back-with-score				
Tags	**Hindi**	**Angika**	**Magahi**	**Bhojpuri**	**AVG**	**Hindi**	**Angika**	**Magahi**	**Bhojpuri**	**AVG**
ADJ	0.89	0.68	0.74	0.76	0.73	0.95	0.73	0.74	0.76	**0.74**
ADP	0.96	0.82	0.94	0.94	**0.90**	0.98	0.82	0.94	0.94	**0.90**
ADV	0.95	0.36	0.51	0.48	**0.45**	0.95	0.36	0.49	0.48	0.44
AUX	0.96	0.78	0.84	0.81	**0.81**	0.98	0.78	0.84	0.82	**0.81**
CCONJ	0.93	0.70	0.87	0.88	0.82	0.97	0.89	0.87	0.88	**0.88**
DET	0.8	0.44	0.47	0.59	0.50	0.81	0.43	0.58	0.55	**0.52**
NOUN	0.93	0.83	0.85	0.87	**0.85**	0.95	0.83	0.85	0.87	**0.85**
NUM	0.9	0.75	0.69	0.74	0.73	0.92	0.75	0.69	0.82	**0.75**
PART	0.91	0.47	0.79	0.75	**0.67**	0.88	0.44	0.79	0.74	0.66
PRON	0.93	0.69	0.7	0.63	0.67	0.96	0.78	0.7	0.75	**0.74**
PROPN	0.85	0.5	0.57	0.49	0.52	0.87	0.64	0.57	0.59	**0.60**
PUNCT	0.98	0.99	0.98	1.00	**0.99**	0.98	0.99	0.98	1.00	**0.99**
SCONJ	0.83	0.63	0.68	0.63	**0.65**	0.83	0.63	0.68	0.63	**0.65**
SYM	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
VERB	0.95	0.71	0.75	0.82	**0.76**	0.97	0.74	0.75	0.82	**0.76**

Table 3: Tag-wise F1-scores using MuRIL for POS tags across all languages for the zero-shot baseline ('Hindi FT') and 'look-back-with-score' method. Average (AVG) is calculated using Angika, Magahi, and Bhojpuri scores.

Recall to Creation: Generating Follow-Up Questions Using **Bloom's Taxonomy** and **Grice's Maxims**

Archana Yadav, Harshvivek Kashid, Medchalimi Sruthi, B JayaPrakash, Chintanapalli Raja Kulayappa, Mandala Jagadeesh Reddy, Pushpak Bhattacharyya, *From Recall to Creation: Generating Follow-Up Questions Using Bloom's Taxonomy and Grice's Maxims*, ACL-Industry (Oral) 2025. Vienna, Austria, July 27-August 1, 2025.

AI in Cars

- AI makes in-car assistants smarter and more responsive.
- Learns driver preferences to suggest routes, recommend services, and manage entertainment.
- Provides intuitive interfaces for seamless control of vehicle features and information.
- Integrates with the infotainment system to manage functions and answer questions about the car.
- Helps with navigation—offering directions, alternate routes, and points of interest.

IIT Bombay & Hyundai Collaboration

Hyundai's in-car AI assistant understands user queries and enhances the driving experience.

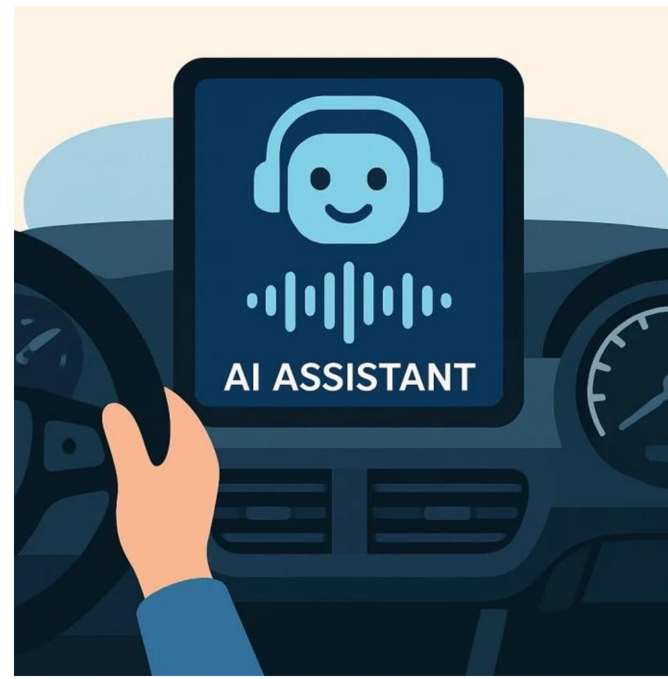
Problems to be addressed:

- ❑ It often falters on multi-turn dialogues and complex follow-up questions.
- ❑ Manual testing inadequate: is slow, resource-intensive, and hard to scale, so an automated evaluation system is needed.
- ❑ Coverage: Evaluation must span domains—navigation, weather, entertainment, vehicle controls/features—especially in the Indian context.

Interaction between Car Driver and In-Car AI assistant

Asks Questions

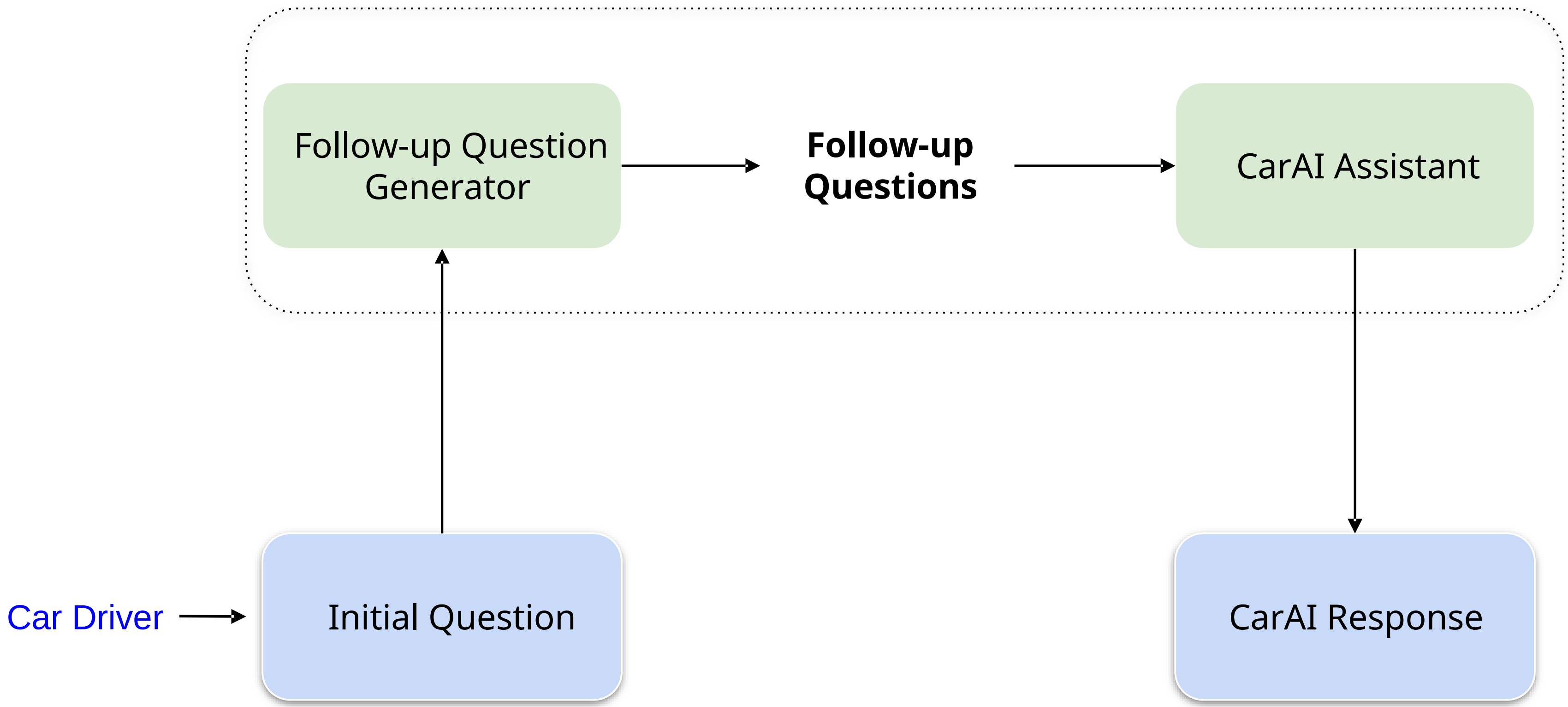
Responds to Queries



3Vs of CarAI Testing: Volume, Variability and Velocity

- **Volume**: Generate large numbers of follow-up questions to cover all driving scenarios.
- **Variability**: Include diverse levels of cognitive complexity and domains.
- **Velocity**: Use scalable, automated frameworks for rapid test-case generation and evaluation.

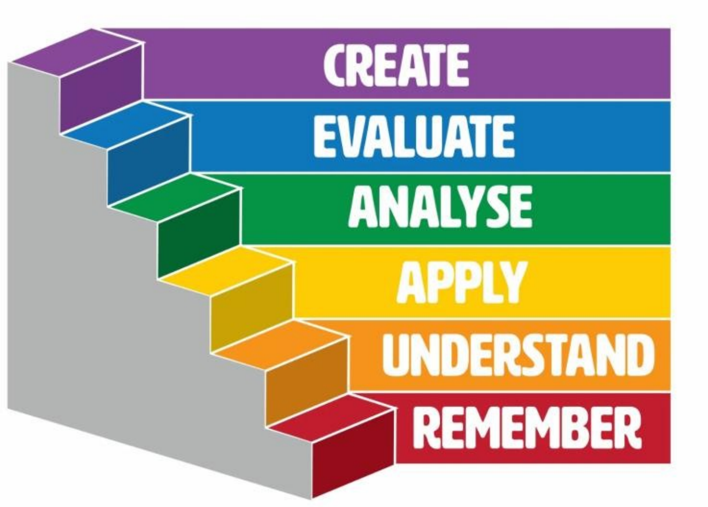
Proposed System Preview



Foundations

Bloom’s Taxonomy

What is Bloom's Taxonomy???



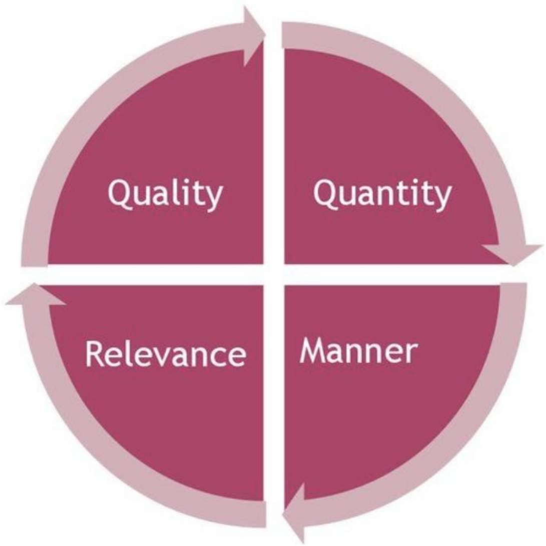
- Bloom's taxonomy was developed to provide a common language for teachers to discuss and exchange learning and assessment methods.
- Originally proposed in *1956* by Benjamin Bloom, the taxonomy has been updated to include the following **six levels** of learning: **Remember, Understand, Apply, Analyze, Evaluate and Create.**

Bloom’s Taxonomy guided Follow-up Questions

Level 1 - Initial Query	**What is photosynthesis?**
Level 2 - Understand	In your own words, explain how plants use sunlight to make food.
Level 3 - Apply	Given a sealed jar with a water plant under sunlight, predict what would happen to the oxygen level in the jar over time.
Level 4 - Analyze	Compare and contrast the roles of chlorophyll and stomata in the photosynthesis process.
Level 5 - Evaluate	“Plants perform photosynthesis only during the daytime.” Support your judgment with evidence.
Level 6 - Create	Design a simple experiment to test how different colors of light affect the rate of photosynthesis in spinach leaves.

Bloom’s Taxonomy guided Follow-up Questions in CarAI

Level 1 - Initial Query	**How do I make a call?**
Level 2 - Follow-up Question 1 (understand)	What are the different options I have to make a call in this car?
Level 3 - Follow-up Question 2 (apply)	How does the call-making process differ from my previous car model?
Level 4 - Follow-up Question 3 (analyse)	What are the advantages of using the car's built-in calling system over my phone's calling feature?
Level 5 - Follow-up Question 4 (evaluate)	Can you explain how the car's calling system integrates with my phone's contact list and how it affects call quality?
Level 6 - Follow-up Question 5 (create)	How can I use the call-making feature in this car to improve my safety while driving, such as by using voice commands or hands-free modes?



Grice's Conversational Maxims

Grice's Maxims

The four maxims are:

- The **maxim of quality**, where one tries to be truthful, and does not give information that is false or that is not supported by evidence.
- The **maxim of quantity,** where one tries to be as informative as one possibly can, and gives as much information as is needed, and no more.
- The **maxim of relation**, where one tries to be relevant, and says things that are pertinent to the discussion.
- The **maxim of manner**, when one tries to be as clear, as brief, and as orderly as one can in what one says, and where one avoids obscurity and ambiguity.

Maxim of Quality

The Maxim of Quality ensures truthfulness and consistency.

Initial Question:	What is the current battery health status?	
Good Follow-up Question:	Based on the last diagnostic report, has the battery health improved or declined?	Consistent with previous info, logically follows up on the status
Bad Follow-up Question:	If the battery is at 100%, why is it showing as completely dead?	Contradicts known facts, causing confusion and illogical inference

Maxim of Quantity

Follow-up questions should introduce an appropriate amount of new information, neither too broad nor too narrow, so the AI can respond effectively and informatively.

Initial Question:	How do I use voice commands?	
Good Follow-up Question:	Which voice commands help me navigate to nearby restaurants and control music playback?	Provides a clear, focused scope with manageable new info.
Bad Follow-up Question:	Tell me everything about all the voice commands your system supports.	Overwhelming with too much new information.

Maxim of Relation

Follow-up questions should be directly related to the previous topic, maintaining a coherent and purposeful conversation flow.

Initial Question:	What’s the current tire pressure?	
Good Follow-up Question:	What is the recommended tire pressure for highway driving?	Relevant and on-topic.
Bad Follow-up Question:	Can you tell me the weather forecast for tomorrow?	Irrelevant; breaks conversational flow.

Maxim of Manner

Follow-up questions should be straightforward and brief, avoiding unnecessary words or confusing phrasing.

Initial Question:	How do I connect my phone to the car’s Bluetooth?	
Good Follow-up Question:	What’s the quickest way to pair my phone with Bluetooth?	Short and to the point
Bad Follow-up Question:	Can you please explain in detail the entire process of connecting my mobile device to the vehicle’s Bluetooth system step-by-step?	Too long and complicated

Gricean Principles for Evaluating CarAI Dialogue

- **Evaluate follow-up questions** from a driver's perspective.
  - **Quality:** Is the question logically consistent?

$$\mathrm{LC}(Q_i \mid C_i) \;=\; \mathrm{Entail}_{\texttt{roberta}}(Q_i, C_i)$$

  - **Quantity:** Does the question introduce the right amount of new information?

$$H(Q_i \mid C_i) = -\sum_{w \in Q_i} P(w \mid C_i) \log P(w \mid C_i)$$

  - **Relation:** Is the question relevant to the ongoing topic?

$$\text{Relevance Score}(Q_i, C_i) = \cos(v(Q_i), v(C_i))$$

  - **Manner:** Is it clear, concise and easy to understand?

$$\text{Clarity}(Q_i) = \frac{1}{1 + \mathsf{ADD}(Q_i)}$$

ADD(): how far are the words from their heads in the DT

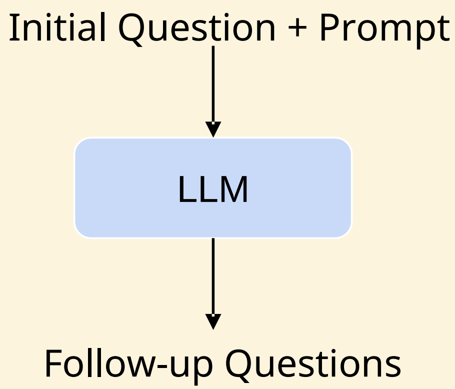
JMP

The System

Core Tasks: Generation & Evaluation

Task 1: Follow-up Question Generation

- ❑ Initial question annotation
- ❑ Follow-up question generation based on Bloom's Taxonomy



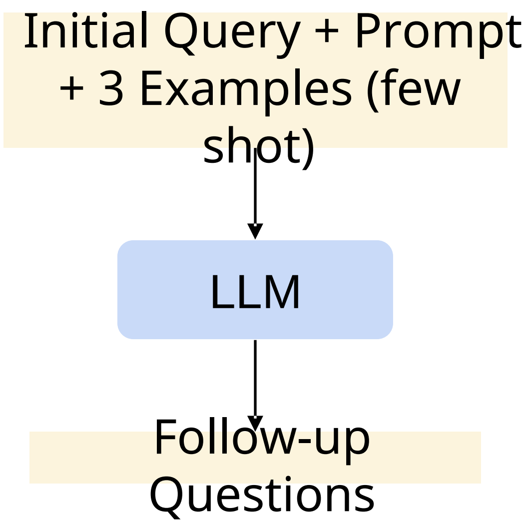
Task 2: Evaluation of Follow-up Questions

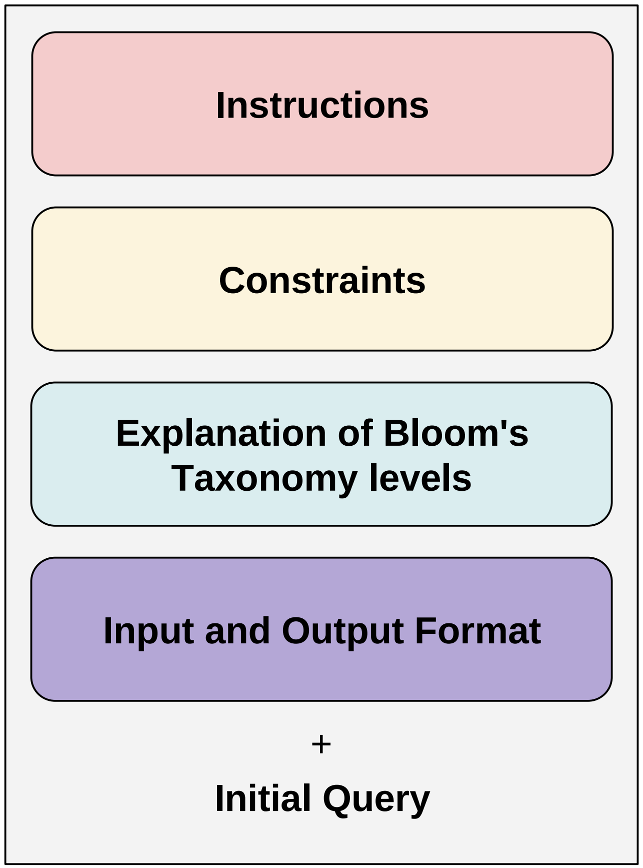
- ❑ Formulation of evaluation metrics based on Grice's Maxims
- ❑ Human evaluation guidelines
- ❑ Human and Grice's Maxims based Evaluation on four metrics:
  - ❑ Logical Consistency
  - ❑ Informativeness
  - ❑ Relevance
  - ❑ Clarity

Methodology

- ❑ Few-shot prompting: three human-annotated in-context examples covering diverse in-car AI assistant domains

- LLMs used for generation:
  - GPT4
- LLMs used for generation:
  - ❑ Qwen-7B-Instruct
  - ❑ Mistral-7B-Instruct
  - ❑ OLMoE-1B-7B-Instruct
  - ❑ Llama-3.1-8B-Instruct





Prompt Template

System Prompt

```
You are an AI tasked with generating follow-up questions for a car driver to ask an in-car AI assistant. The questions will assess the AI's understanding of the car's features and de
Constraints:
Feature Neutrality: Do not assume, add, or imply any car features that are not explicitly mentioned or suggested in the seed question. Base all follow-up questions solely on the con
Answer-Agnostic: Focus on the driver's interaction with the car and how the car's features enhance the driving experience without delving into internal technical details or making a
Driver-Focused Interaction: Ensure that all questions center on the driver's use and experience with the car. Do not include questions regarding the car's internal mechanisms, data-
Single-Faceted: Each question must target a single concept or action to maintain clarity. Avoid compound or multi-part questions.
Sequential Progression: The follow-up questions should build upon each other, moving from basic recall (Level 1) to more advanced cognitive tasks (Level 6).
Bloom’s Levels Only: Only generate questions for Levels 2 through 6 of Bloom's Revised Taxonomy. Do not introduce any levels beyond Level 6.
Avoid Unrelated Topics: Refrain from generating questions about topics such as software details, UI design, or interface functionalities that fall outside the scope of the in-car as

Explanation of Bloom's Revised Taxonomy Levels:
Level 1 (Remember): Involves recalling or recognizing facts and basic concepts. (This level is provided as the seed question.)
Level 2 (Understand): Involves explaining ideas or concepts. Questions at this level ask for clarification or interpretation.
Level 3 (Apply): Involves using information in new or concrete situations. Questions should prompt practical use or demonstration of how a feature could be used.
Level 4 (Analyze): Involves breaking information into parts and exploring relationships. Questions should prompt examination of reasons, causes, or underlying structures.
Level 5 (Evaluate): Involves making judgments based on criteria and standards. Questions should encourage assessment or justification of decisions.
Level 6 (Create): Involves putting elements together to form a new, coherent whole or propose alternative solutions. Questions should prompt the generation of original ideas or new 

Output Instructions:
    - Output only the five generated follow-up questions.
    - Do NOT include any explanations, extra text, or formatting.

Input Format:
<seed>seed_question_str</seed>

Output Format:
<question>question_1_str</question>
<question>question_2_str</question>
<question>question_3_str</question>
<question>question_4_str</question>
<question>question_5_str</question>

Instruction: Output only five lines each corresponding to a question from level 2 to level 6 as described before, and nothing else. Donot provide any additional explanation or reaso
```

Human Evaluation OF FQG

Scoring Definitions (1-5 Likert Scale):

- **1 (Very Poor)** - The metric is not followed at all while generating the follow-up question based on the previous questions.
- **2 (Poor)** - The metric is followed only to a limited extent while generating the follow-up question based on the previous questions.
- **3 (Fair)** - The metric is followed to a good extent while generating the follow-up question based on the previous questions.
- **4 (Good)** - The metric is followed mostly while generating the follow-up question based on the previous questions.
- **5 (Excellent)** - The metric is followed completely while generating the follow-up question based on the previous questions.

Results

Question Generation Model	Logical Consistency	Informativeness	Relevance	Clarity
Qwen-7B-Instruct	4.70	**4.70**	4.63	**4.79**
Mistral-7B-Instruct	**4.79**	4.65	**4.66**	4.78
OLMoE-1B-7B-Instruct	4.68	4.63	4.46	4.75
Llama-3.1-8B-Instruct	4.62	4.44	4.33	4.63

Human evaluation scores (on a 5-point likert scale, ranging from 1 to 5) for follow-up questions generated by four models, Qwen-7B-Instruct, Mistral-7B-Instruct, OLMoE-1B-7B-Instruct, and Llama-3.1-8B-Instruct, across four metrics: Logical Consistency, Informativeness, Relevance, and Clarity.

ANALYSIS

1. Logical Consistency (Maxim of Quality)

Evaluated using entailment from roberta-large-mnli

Initial Question:
"How do I turn on the AC in this car?"

Follow-up Question	**Consistency Score**
"Can I also adjust the fan speed manually?"	5
"Does this car support autopilot like a Tesla?"	1

ANALYSIS

2. Informativeness (Maxim of Quantity)

Evaluated using Conditional Entropy

Initial Question:
"Can you tell me about the car’s mileage?"

Follow-up Question	**Informativeness Score**
"What’s the highway mileage compared to city mileage?"	5
"Can you tell me about the car’s mileage again?"	1

ANALYSIS

3. Relevance (Maxim of Relation)

Evaluated via cosine similarity between follow-up and context embeddings

Initial Question:
"How do I connect my phone to Bluetooth?"

Follow-up Question	Relevance Score
"Can I also stream music once it’s connected?"	5
"What's the ground clearance of this car?"	1

ANALYSIS

4. Clarity (Maxim of Manner)

Evaluated using Average Dependency Distance (ADD); lower is clearer

Initial Question:
"How do I open the sunroof?"

Follow-up Question	**Clarity Score**
"Can I open it halfway?"	5
"Is it within the realm of possibility for the sunroof's current configuration to be partially retracted using available controls?"	1

Results

Question Generation Models	Techniques	Logical Consistency	Informativeness	Relevance	Clarity
Qwen-7B-Instruct	Few-shot	**0.9122**	0.5108	**0.6025**	**0.2743**
	Zero-shot	0.8762	0.4897	0.5613	0.2582
Mistral-7B-Instruct	Few-shot	<u>0.9052</u>	**0.5991**	<u>0.5917</u>	<u>0.2723</u>
	Zero-shot	0.8830	0.5035	0.5715	0.2684
OLMoE-1B-7B-Instruct	Few-shot	0.8720	0.4693	0.5569	0.2688
	Zero-shot	0.8486	0.4627	0.5453	0.2633
Llama-3.1-8B-Instruct	Few-shot	0.8893	<u>0.5906</u>	0.5559	0.2600
	Zero-shot	0.8847	0.5098	0.5694	0.2564

Evaluation of Follow-up Question Generation Models on four metrics based on the GriceWise evaluation framework. The best scores are bolded, and the second-best scores are underlined.

LLM-based Evaluation (Along with Correlation with Human Evaluation)

Evaluation Method	Logical Consistency (Maxim of Quality)		Informativeness (Maxim of Quantity)		Relevance (Maxim of Relation)		Clarity (Maxim of Manner)	
	ρ	τ	ρ	τ	ρ	τ	ρ	τ
GriceWise Evaluation	0.57	0.48	0.56	0.47	0.61	0.52	0.72	0.60
LLM-based Evaluation	0.63	0.62	0.65	0.63	0.66	0.64	0.76	0.73

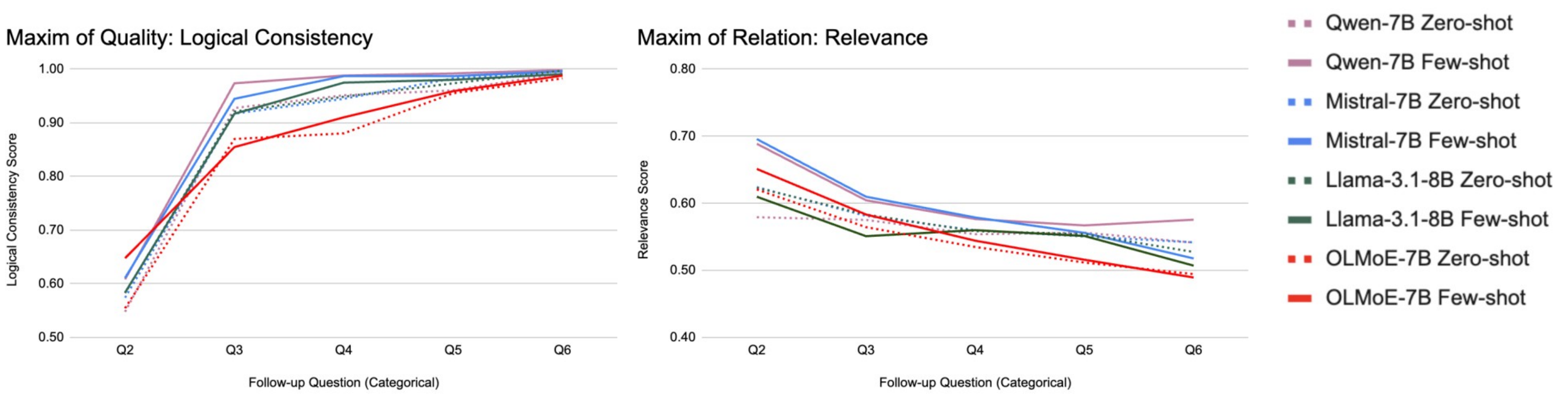
Spearman's ρ and Kendall's τ correlation with human evaluation of GriceWise Evaluation and LLM-based evaluation across four metrics. *gpt-4o-mini* was used for the LLM-based evaluation.

LLM-based evaluation has shown better correlation with human evaluation than the GriceWise formula-based evaluation on all the four metrics.

Qualitative Insights on GriceWise Metric Trends

Logical Consistency (Maxim of Quality): Sharp increase from Q2 to Q4, then plateaus; few-shot > zero-shot. The GriceWise evaluation for logical consistency is binary (0 or 1), so the sharp increase reflects a growing number of responses being judged fully consistent as the model gains context.

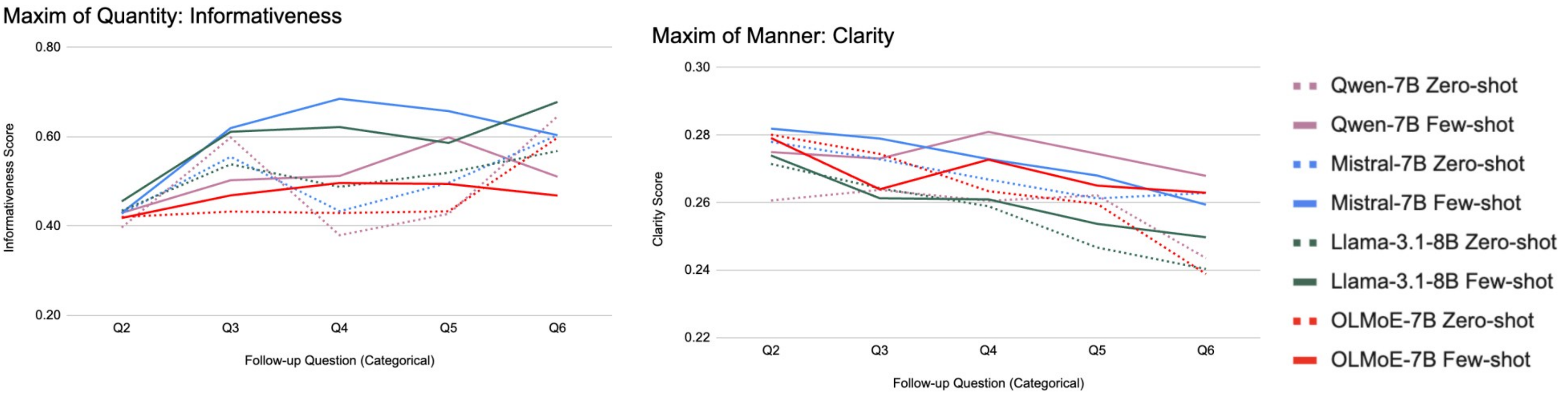
Relevance (Maxim of Relation): Relevance gradually decreases from Q2 to Q6 as questions grow more abstract and harder to align with the main topic; few-shot prompting offers some improvement by providing better grounding, but cannot fully prevent the decline.



Qualitative Insights on GriceWise Metric Trends

Informativeness (Maxim of Quantity): Gradual improvement across Q1 to Q5. Few-shot prompting provides better guidance, yielding richer follow-up questions.

Clarity (Maxim of Manner): Clarity declines steadily as question chains grow longer, often introducing verbosity or ambiguity; few-shot examples help maintain concise and direct phrasing, mitigating this effect.



Conclusions, Future

- NLP= Language+Computation→ Linguistics+Probability
- Rare language phenomena like sarcasm, metaphor, hyperbole need to be detected and generated: *imp for conversational AI agents; has direct impact on business interest*
- Low resource setting in important: ELRLs, AI Agents